Flash loan — , in one atomic transaction

Start with **$0**. Borrow a pile, arbitrage the price gap between two DEXes, repay in the **same tx**. Repay covered → keep the profit. Not covered → the whole tx **reverts**, you lose nothing but gas.

Flash-borrow amount **4,000** tUSDC · fee 0.09% = **3.6**

Execute flash loan (1 transaction)

1
Flash pool
borrow 4,000

2
DEX-cheap · buy MAX
@ 1900

3
DEX-dear · sell MAX
@ 2100

4
Repay pool
repay 4,004

5
Profit (yours)
+214

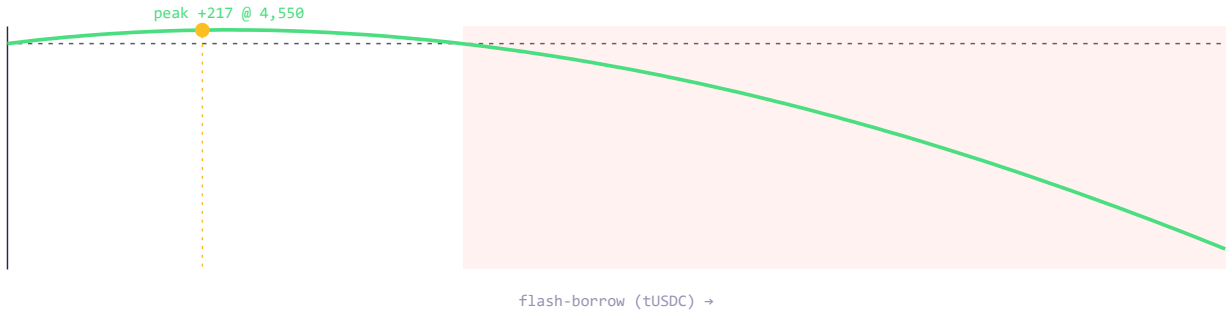
Borrowed	MAX bought (cheap)
4,000	2.056 MAX
tUSDC from selling (dear)	Must repay
4,218	4,004

✓ TX WOULD SUCCEED — profit +214 tUSDC from $0 capital

repay 4,004 ≤ 4,218 you got from the arb

Profit vs borrow size — borrow too much and **your own trades eat the gap** (price impact) → you can't repay → revert zone

profit revert zone (profit < 0) your current borrow



Zero capital. You never owned the borrowed tUSDC — you got it and gave it back inside one transaction. No collateral, because there is nothing to seize: if you don't repay by the last line of the tx, the chain **rewinds the whole thing** as if it never ran.

All-or-nothing (atomicity). Borrow, buy, sell, repay are one indivisible action. Profit ≥ 0 → it commits, you keep the leftover. Profit < 0 → the repay line fails → **everything undoes**. Your only real cost is gas. No holding, no HF, no keeper can catch you — there is no "position" to catch.

Why more borrow ≠ more profit. Buying MAX on the cheap DEX pushes its price **up**; selling on the dear DEX pushes its price **down**. Your own trades close the very gap you're eating (price impact / §08). Past the optimum, the gap is gone and fees dominate → profit turns negative → revert. **The arb is self-limiting; the pool's depth caps your take.**

Why this can't be done by hand. Manually you'd need three separate transactions — and after step 1 you'd be holding an **uncollateralized** loan the chain would never have given you. Atomicity only exists **inside** one tx, so a flash loan **must** be a contract. This is the door from "click by hand" to "write a program."

MB/CSE 599 · AI Finance · flash loan = zero-capital atomic arbitrage · next: the real one-click flash-loan contract